Deepak Mallampati

APPLIED AI ENGINEER - LLM & FULL-STACK AI PLATFORMS

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US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years designing, building, and shipping AI-powered platforms and intelligent automation into production. Builds full-stack AI systems end-to-end across backend services, APIs, and frontends (Python, SQL, React) powered by Claude and other LLMs. Delivered agentic orchestration that cut token cost by 42%, RAG pipelines that reduced hallucination by 40% at 92% precision, and automation infrastructure that lowered failed production runs by 60% - owning problems from discovery through rollout.

CORE COMPETENCIES

AI-Powered Platforms • LLMs (Claude, GPT-4o, Gemini) • Python & SQL • React / Full-Stack • Intelligent Automation & Orchestration • RAG & Vector Search • Prompt Engineering • AWS / Docker / Kubernetes • Distributed Systems

Generative AI & LLMs: Claude Sonnet, GPT-4o/GPT-5, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT)

Languages & Frameworks: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Agile/Scrum

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | CI/CD | MLflow, W&B | Observability

Data: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present | USA*

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Engineered high-performance **RAG pipelines** over large document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination on domain-specific queries.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark model iterations before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard *Jan 2024 - Jan 2025 | USA*

AI Engineer Intern

- Developed AI agents and backend services for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered content controls for secure, policy-compliant AI responses, reducing unsafe outputs by 42% while maintaining quality.
- Partnered with product, platform, and data teams to deliver LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University *Oct 2023 - Jan 2024 | USA*

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 for controllable long-form generation; authored an IEEE-format conference paper.

Emerson *Jun 2022 - Apr 2023 | India*

ML Trainee

- Trained supervised regression/classification models for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving 89% F1 on custom NER.

PROJECTS

MangaLens - shipped Chrome extension (React + LLM)

Live full-stack AI product delivering real-time translation of manga/webtoons across 29+ sites - front-end React UI plus LLM translation backend, turning a multi-step manual chore into a single inline action.

career-ops - autonomous AI job-search pipeline

AI-driven system that scans listings, scores fit, and generates tailored applications overnight - full-stack automation and orchestration with direct practical impact.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive data without exposing identities, quantifying the privacy-utility trade-off.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025 | GPA 3.70/4.0*